

CoCoCorpus annotation guideline

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1 Introduction

Inherently polysemous nouns, also called dot-type nouns, have multiple meaning facets that can be targeted at the same time. This phenomenon is referred to as copredication. For example, in example (1-a), the polysemous noun *book* has an info facet and an event facet. The verb *memorized* targets the info facet and the adjective *heavy* targets the event facet. On the other hand, coercion is the phenomenon where the predicate can also target a facet that is not available in the noun, like in example (1-b), *finished* targeted the event-facet, which is not present in the meaning of *book*.

- (1) a. He *memorized* the *heavy* **book**.
- b. She *finished* the **book**.

Copredication and coercion have been extensively studied in linguistics but lack annotated data. This annotation project aims to provide data for empirical and quantitative research on these phenomena.

We conduct a semi-automatic annotation on sentences from selected corpora. Currently the sentences are from BookCorpus(Zhu et al., 2015) and ACL Anthology Corpus(Bird et al., 2008). The sentences containing a multiple predication on selected dot-type nouns are extracted from the corpora, parsed, and automatically annotated using the classifier from Yavas et al. (2023). The automatic annotation is then revised by at least two human annotators, and the gold standard annotation is acquired after adjudication.

2 Facet selection

2.1 General principle

In our annotation task, facet selection is distinguished from coercion. This part introduces the annotation of facet selection.

Currently in our task, the nouns are preselected, so the facets of the nouns are known. Therefore, the main task in the annotation of facet selection is to decide whether the predicate targets a certain facet. To achieve this, a practical way is to replace the dot-type nouns with other simple type nouns

and check the felicity of the replaced phrase. Taking the phrase *burn the book* as an example, one can try to replace the dot-type noun *book* with physical object nouns such as *tree* and *table*, or info nouns such as *poem* and *message*. The phrase ‘burn the tree/table’ is felicitous but ‘#burn the poem/message’ is not acceptable, therefore ‘burn’ targets the object facet instead of the info facet. The possibility of predication is also related to other factors such as the detailed content meaning of the words, so it is necessary to try more nouns to make sure the predicate can or cannot target a certain meaning facet.

Some predicates are able to target different facets depending on the context. In these cases, the annotator should also take the context into consideration and choose the most possible targeted facet. For example, in (2-a), *first* targets the object facet of *book*, while in (2-b), *first* targets the event facet.

- (2) a. I read the *first* **book** on the shelf.
- b. I read the *first* **book** he wrote.

2.2 Object facet

The object facet selection is labelled as ARTIFACT in our annotation. This label, as well as the others, comes from T-PAS taxonomy (Jezek et al., 2014), from which our automatic classifier has been trained on.

Common dot-types that involve object facets include nouns of type *food • event* (e.g. *lunch*, *meal*), *object • info* (e.g. *book*, *diary*), *object • location* (e.g. *tower*, *classroom*). Typical predicates that target object facets include *burn*, *throw*, and *red*. (3) are two examples from BookCorpus where the predicates *bring* and *thick* target the object facets of the dot-type nouns.

- (3) a. Go *bring* your sister a late **lunch**.
- b. Beside her, Jacob was pouring over a *thick* **textbook**, his face scrunched up in confusion.

Digital objects are also treated as having object facets. For example, in (4), the relation between *downloaded* and *book* is annotated as ARTIFACT.

- (4) He is *downloading* the interesting **book**.

2.3 Event facet

The event facet selection is labelled as EVENT. Common dot-types that involve event facets include nouns of type *food • event* (e.g. *lunch*, *meal*) and *info • event* (e.g. *speech*, *lecture*). Typical predicates that target event facets include *organize*, *schedule*, and *early*. (5) are two examples from BookCorpus where the predicates *quick* and *carry on* target the event facets of the dot-type nouns.

- (5) a. In the morning he ate a *quick* **breakfast** in his room, as his staff prepared for an immediate return.
- b. I have lost the ability to *carry on* a simple **conversation** without tripping over my words.

Besides typical events, sound is also treated as events in our annotation, although some ontological research suggests that it is different from either physical objects or events (Dokic, 2007). The reason for this treatment is related to the predication about sounds. For verbs and adjectives that predicate over “sound nouns” like *noise* and *loud*, it is usually natural for them to also predicate over events, but not so natural to predicate over physical objects. For example, *hear a noise* and *hear the dinner* are both acceptable, but *#hear a pizza* is rather strange. Also *a loud noise* and *a loud dinner* are acceptable, but not *#a loud pizza*.¹ Therefore, for verbs and adjectives that address the sound, we annotate them as EVENT if the event facet is available in the noun.

When predicating over *food • event* nouns, ordering adjectives usually target the event facet, as in (6). This is because in normal contexts, the meals are usually ordered by the time they are eaten instead of the order of the physical food. Similarly, as introduced above, most ordering adjectives target the event facet of *event • info* nouns.

- (6) I eat my *first* unhurried, no nagging, peaceful **meal** in that time.

2.4 Info facet

The info facet selection is labelled as INFO. Common dot-types that involve info facets include nouns of type *obj • info* (e.g. *textbook*, *diary*) and *event • info* (e.g. *speech*, *lecture*). Typical predicates that target object facets include *memorize*, *recite*, and *informative*. (7) are two examples from BookCorpus where the predicates *overheard* and *classic* targets the info facets of the dot-type nouns.

- (7) a. the very next morning i happened to be walking to my office when i *overheard* a heated **conversation** in the hallway by two of our custodians.
 b. Ronnie personally liked seeing the *classic* **novel** set in high school; she related to the characters better and enjoyed that it was about more than just finding a boyfriend.

2.5 Object facet & Event facet

In some cases, the object facet and the event facet of a *food • event* noun are both targeted by a predicate. These cases are annotated as ARTIFACT_EVENT. (8) are some examples.

- (8) a. “We just *want* a warm **meal** and a bed for the night is all”, the first companion reaffirms.
 b. No more than a dozen people sitting at tables, *grabbing* an early midday **meal**.

¹Phrases like *hear the water* and *a loud washing machine* are acceptable but they involve coercion, because *hear the water* means to hear the flowing/bubbling of the water, and *a loud washing machine* means a washing machine that is loud when operating.

- c. Once past the initial security checkpoint it occurred to Jane that she could not *remember* her last **meal**, but there were lines outside all the food stalls.
- d. I am planning a *big meal* and I expect you here with the family.

These cases involve the following predicates:

- ‘Wanting’ verbs: *want, crave, need*, etc. Usually, the person wants both the food and the eating event, as in (8-a), so the relations between these verbs and *food • event* nouns are annotated as ARTIFACT_EVENT unless the context clearly indicates that only one of the facets is “wanted”.
- ‘Getting’ verbs: *get, fix, grab, receive*, etc. These verbs have relatively light meanings and do not specifically target one of the facets, as in (8-b).
- ‘Memory’ verbs: *remember, forget*, etc. Sometimes the context suggests both facets are targeted, as in (8-c), the person did not remember eating, and also did not remember the food, so both facets are targeted. But if the context suggests one of the facet is preferred, the annotator should only annotated that facet.
- Adjectives that can target many facets: *big, special*, etc. These adjectives can modify different types, and in many cases only one facet is targeted. However, in some situations, multiple facets are targeted, as in (8-d), usually *big meal* means a meal with a lot of food, but “expect you here with the family” suggests that the event is also big, so in this case, both facets are targeted.

2.6 Object facet & Info facet

Some predicates target both the object facet and the info facet of *object • info* nouns and are annotated as ARTIFACT_INFO. These predicates are related to the processing and creation of the *object • info*, such as *read, write, keep (a diary), publish, release*, etc. (9) are some examples where the predicate targets both facets.

- (9) a. I *rewrote* that first **novel**, went onto a second, and then a third.
- b. In three years time when I am *penning* my fifteenth **novel** aboard my shiny, new yacht my opinion on that may well have changed, but in the mean time all I want in return is your opinion.

2.7 Event facet & Info facet

Similar to the cases of *food • event* nouns, “wanting” verbs are also targeting both facets of a *event • info* noun. In these cases, the verbs are annotated as EVENT_INFO. (10) are two examples where both facets are targeted.

- (10) a. This is something that *deserves* a calm **conversation** in the light of day.

- b. Neely *wanted* a long **conversation**, one that covered several issues and many years.

Besides, “hearing” verbs may also target both facets if the context implies that the persons process the content they hear, as in (11).

- (11) Antonia did but only because she wanted to have it out with Brett and didn’t want his other employees to *hear* the **conversation**.

3 Coercion

3.1 General principle

In our project coercion is distinguished from facet selection. Coercion can be identified with the following criteria. First, the facet the predicate targets is obviously not available in the noun, like in (12-a), *begin* targets an event facet but *novel* has no event facets. Second, the phrase cannot be easily transformed into other syntactic constructions. For example, in (12-b), the phrase *earlier summary* cannot be transformed into subject-predicate construction or relative clause, like *#the summary is earlier* or *#the summary that is earlier*. For verb-object coercion, passivization is a useful way to detect coercion. Third, in coercion phrases, the coerced meaning is usually specified by contexts. In (12-a) the coerced event can be reading, writing, or anything related to novels. The context suggests that the coerced event in this sentence should be writing. In (12-b) the coerced event can also be reading, writing or other events. Writing is also more likely according to the context.

- (12) a. In 1997, he wanted to share what the Bible had to say about the end times, so he *began* his first **novel**.
- b. After we chewed on all this for a few minutes, Natalie referenced my *earlier summary* of our interview with Gloria Stallings.

Even if the predicate targets a facet that is available in the noun, it can still be coercion. For example, for the phrase *finish the lunch*, if the context suggests it means finish eating, it is facet selection; if the context suggests it is finishing cooking or other events, it should be labelled as coercion. In (13), *posted* targets an info facet, but according to the context, what the speaker posted is the metadata of the novel (e.g. description or advertisement) instead of the actual content, so this case should be annotated as COERCION_INFO instead of INFO.

- (13) I have *posted* the new **novel** Rome’s Evolution on Amazon, B & N, Kobo and Smashwords.

3.2 Object coercion

The coercion to an object facet, such as *receive a message* and *a delicious picnic*, is labelled as COERCION_OBJ. It is not very common among dot-type nouns.

3.3 Event coercion

The coercion to an event facet is labelled as COERCION_EVENT. It is the most common coercion in our annotation. It usually happens with *enjoy*, phase verbs, or time-related predicates. (14) are some examples of the coercion to the event facet.

- (14) a. Thirteen years ago, when I *finished* my first 180,000 word suspense **novel** and was trying to get an agent, my mother said something to me.
- b. I'll get the chief scientist to check with the atomic energy authority to see if Barclay copies his *weekly* **diary** to anyone there.

3.4 Info coercion

The coercion to an info facet is also less common among dot types. (13) is an example of coercion to the info facet.

4 Other special cases

4.1 Light verb constructions

As observed in previous research, light verbs are different from regular verbs in terms of predicating over dot-type nouns, therefore we decide to annotate light verb constructions as LVB for further research.

We implement the diagnostics from Fleischhauer and Neisani (2020) to distinguish light verb constructions from regular verb predications. The following phrases are considered light verb constructions in our annotation task:

- *have a dinner/breakfast/lunch/meal*
- *have a lecture/conversation/dialogue*
- *give a lecture/speech/summary*
- *deliver a lecture/speech*
- *make a speech/dialogue*

4.2 Delete

Some sentences are suitable for our annotation and needs to be excluded. We use the label DELETE to mark these sentences and delete them during postprocessing. The sentences that should be deleted include:

- The sentence itself is inappropriate, e.g. includes swearing or sexual content.

- There is a parsing error in the sentence and the dot type noun is not targeted by the annotated predicate.
- The target noun in the sentence is of another meaning which is not inherently polysemous.
- The predicate is special and the mechanism of the predication might not be facet selection. These predicates include:
 - *own, nice, fine, good, great, wonderful, bad, awful, amazing, many, more, few, some, entire, such, whole, equal, mean, say, same, certain, particular, own, actual, real, other*

5 Annotation operations

5.1 Annotation platform

The annotation is conducted on the Cococorpus website using Inception platform (Klie et al., 2018). After logging in, the annotator gets into the project page. After clicking on the annotation project, the annotator reaches the webpage shown in Fig 1.

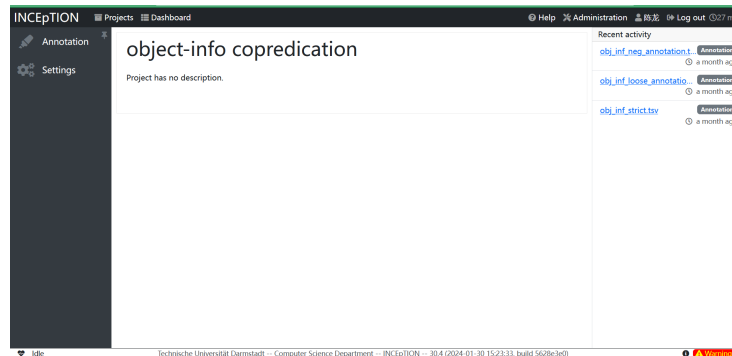


Figure 1: The webpage of the project

The annotator should click on the link for annotation on the left side and reach the page of annotation files (as in Fig 2) to choose the annotated file. After clicking on the chosen file, the annotator will enter the annotation page.

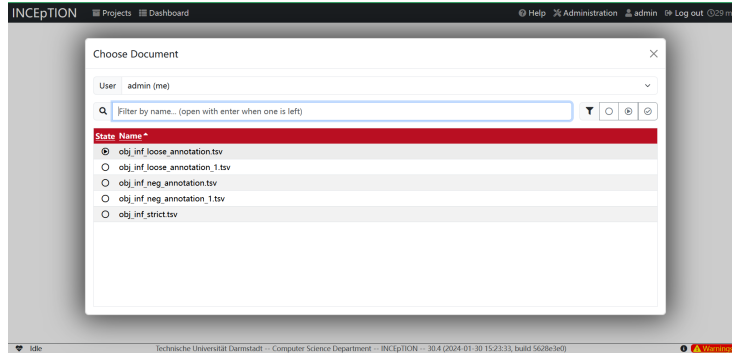


Figure 2: The webpage of annotation files

5.2 Span operations

Currently in our annotation there is no need for any operations on spans. The label on spans denotes the part of speech of the word, and if it is wrong, the sentence has a parsing error and should be deleted. In this case, the annotator should label the edge as DELETE instead of editing the spans.

5.3 Edge operations

The main task of the annotator is to check whether the label on the edge is suitable and make corrections if necessary. To change the label on the edge, the annotator can click on the edge, and select the correct label from the dropdown menu on the right (see Fig 3). If none of the labels from the dropdown menu is suitable, the annotator can also manually type the label in the box.

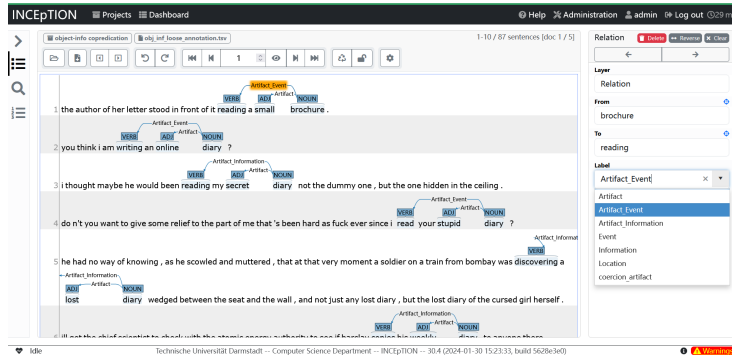


Figure 3: Annotation process

The annotator can also add an edge by toggling the mouse from one span to another. In our task the direction of the edge is defined as from the noun to the predicate.

References

- Bird, S., Dale, R., Dorr, B. J., Gibson, B. R., Joseph, M. T., Kan, M.-Y., Lee, D., Powley, B., Radev, D. R., Tan, Y. F., et al. (2008). The acl anthology reference corpus: A reference dataset for bibliographic research in computational linguistics. In *LREC*.
- Dokic, J. (2007). Two ontologies of sound. *The Monist*, 90(3):391–402.
- Fleischhauer, J. and Neisani, M. (2020). Adverbial and attributive modification of persian separable light verb constructions. *Journal of Linguistics*, 56(1):45–85.
- Ježek, E., Magnini, B., Feltracco, A., Bianchini, A., and Popescu, O. (2014). T-pas: A resource of corpusderived types predicate-argument structures for linguistic analysis and semantic processing. In *Proceedings of LREC*, volume 2014.
- Klie, J.-C., Bugert, M., Boullosa, B., De Castilho, R. E., and Gurevych, I. (2018). The inception platform: Machine-assisted and knowledge-oriented interactive annotation. In *Proceedings of the 27th international conference on computational linguistics: system demonstrations*, pages 5–9.
- Yavas, D. E., Kallmeyer, L., Osswald, R., Ježek, E., Ricchiardi, M., and Chen, L. (2023). Identifying semantic argument types in predication and copredication contexts: A zero-shot cross-lingual approach. In *Proceedings of the 14th International Conference on Recent Advances in Natural Language Processing*, pages 310–320.
- Zhu, Y., Kiros, R., Zemel, R., Salakhutdinov, R., Urtasun, R., Torralba, A., and Fidler, S. (2015). Aligning books and movies: Towards story-like visual explanations by watching movies and reading books. In *Proceedings of the IEEE international conference on computer vision*, pages 19–27.